

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA0706 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Dr.V.PARTHIPAN

Code of Conduct:

I _____ Vandana Lasya(192524186)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

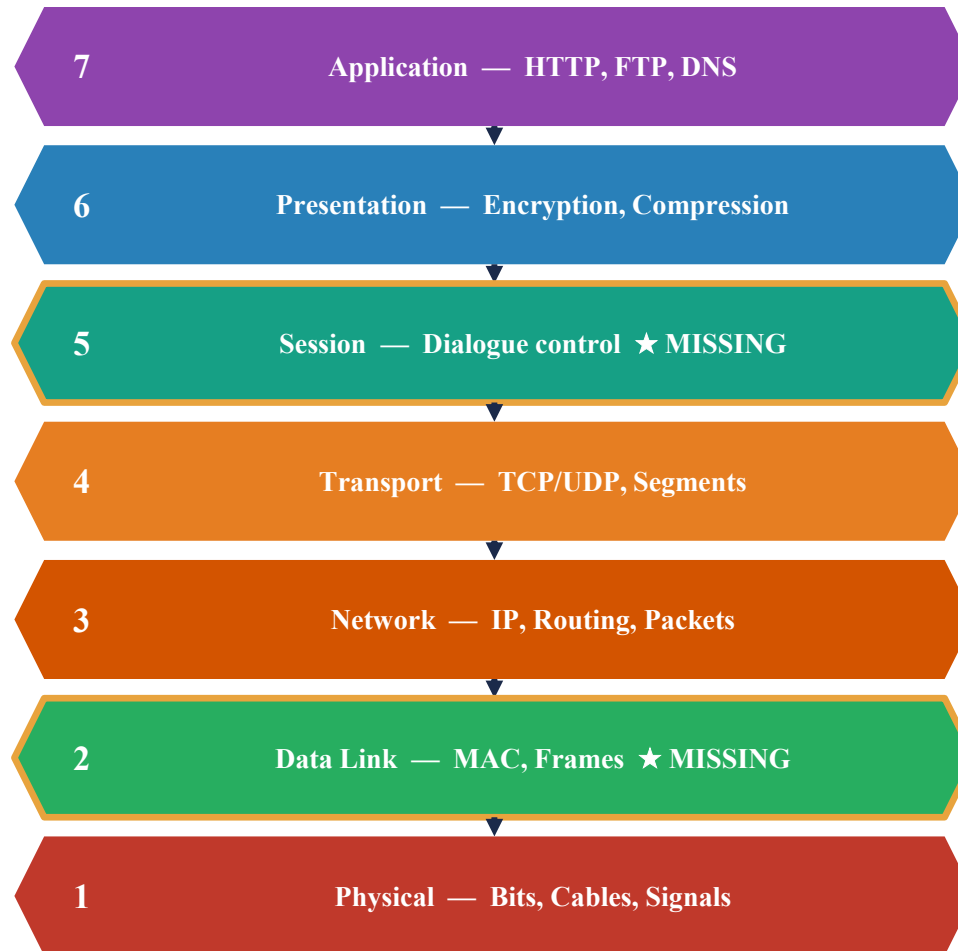
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: VANDANA LASYA

Question 1

Completing the Concept Map of OSI Layers

Application → Presentation → _____ → Transport → Network → _____ → Physical. Shape used: HEXAGON stack



Why essential

- Session: manages dialogue, resumes after interruption
- Data Link: MAC framing + CRC error detection
- Both complete the encapsulation chain end-to-end

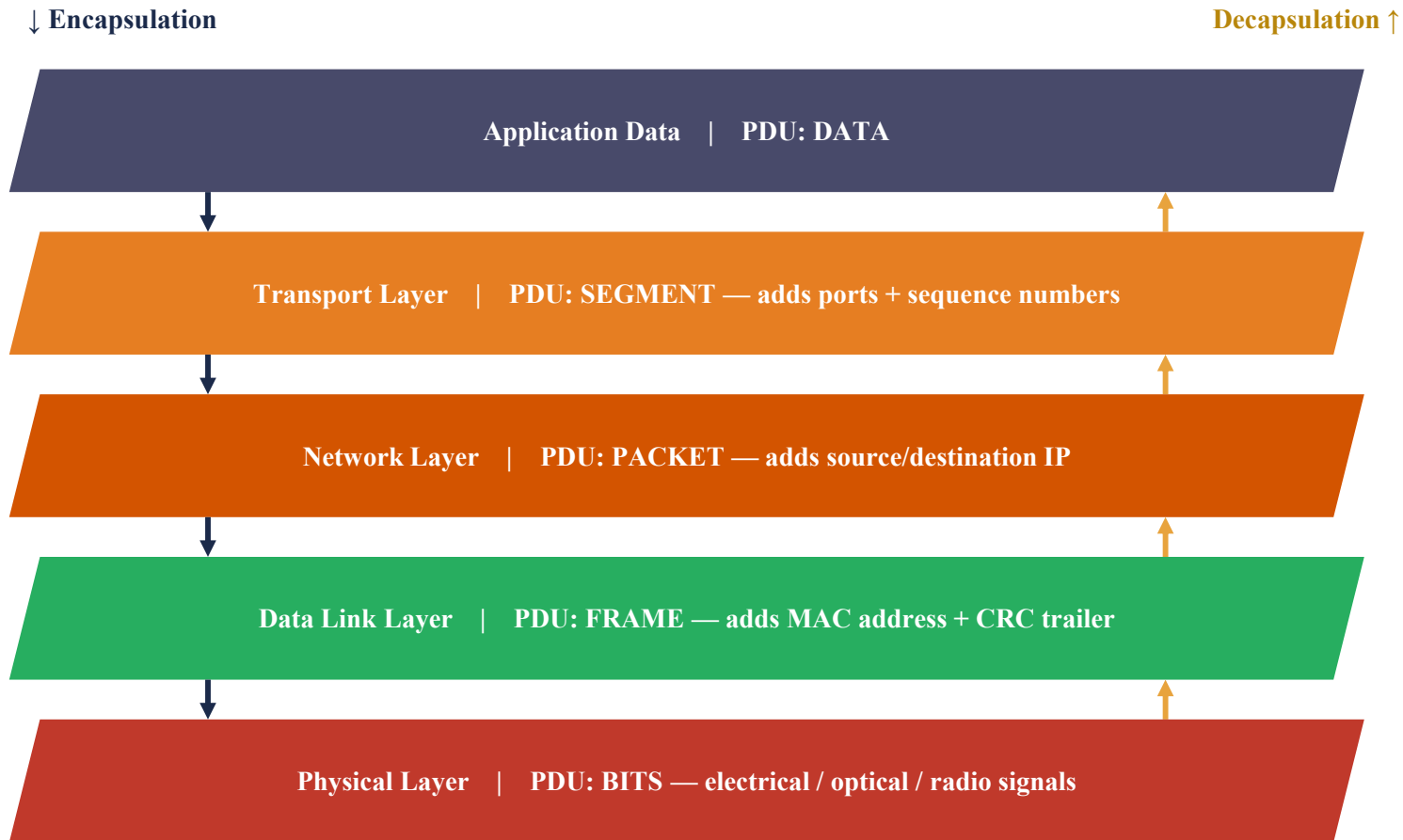
Question 2

Data Units and Their OSI Layers

Each Protocol Data Unit (PDU) mapped to its OSI layer. Shape used: PARALLELOGRAM (data flow)

Key idea

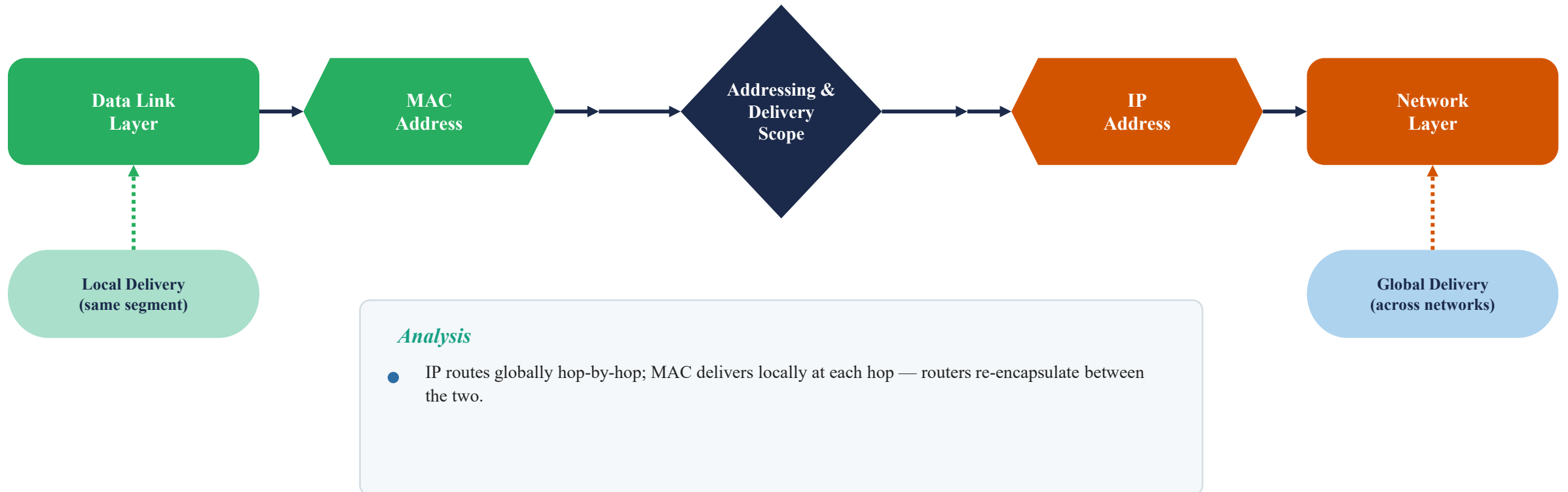
- Encapsulation:
Data→Segment→Packet
→Frame→Bits
- Decapsulation reverses
this at the receiver
- Each layer only reads its
own header



Question 3

MAC Address vs IP Address in Data Delivery

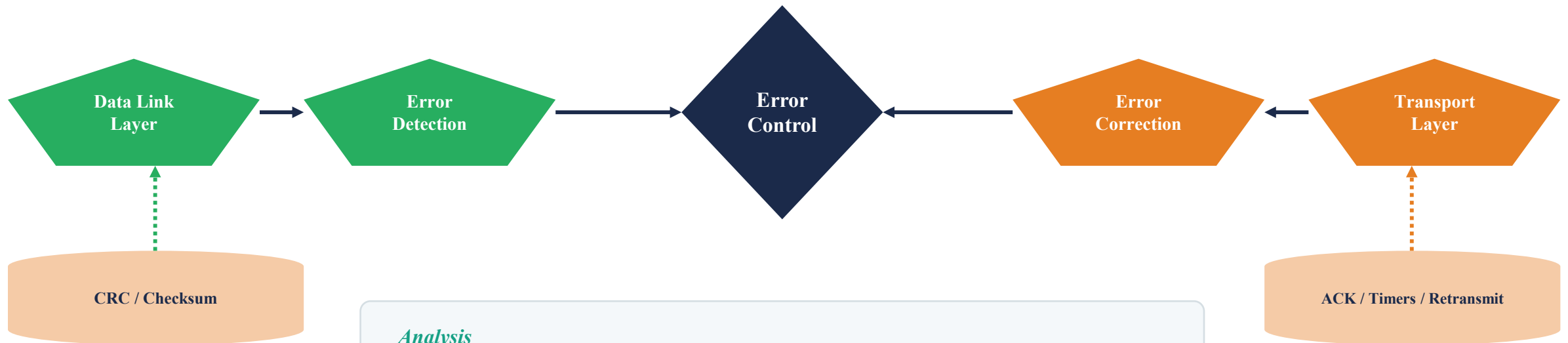
Data Link → MAC Address → Local Delivery | Network → IP Address → Global Delivery | Shapes: OVAL terminators + DIAMOND hub



Question 4

Error Control at Data Link vs Transport Layer

Error Control → Data Link → Error Detection | Error Control → Transport → Error Correction | Shapes: PENTAGON + CYLINDER



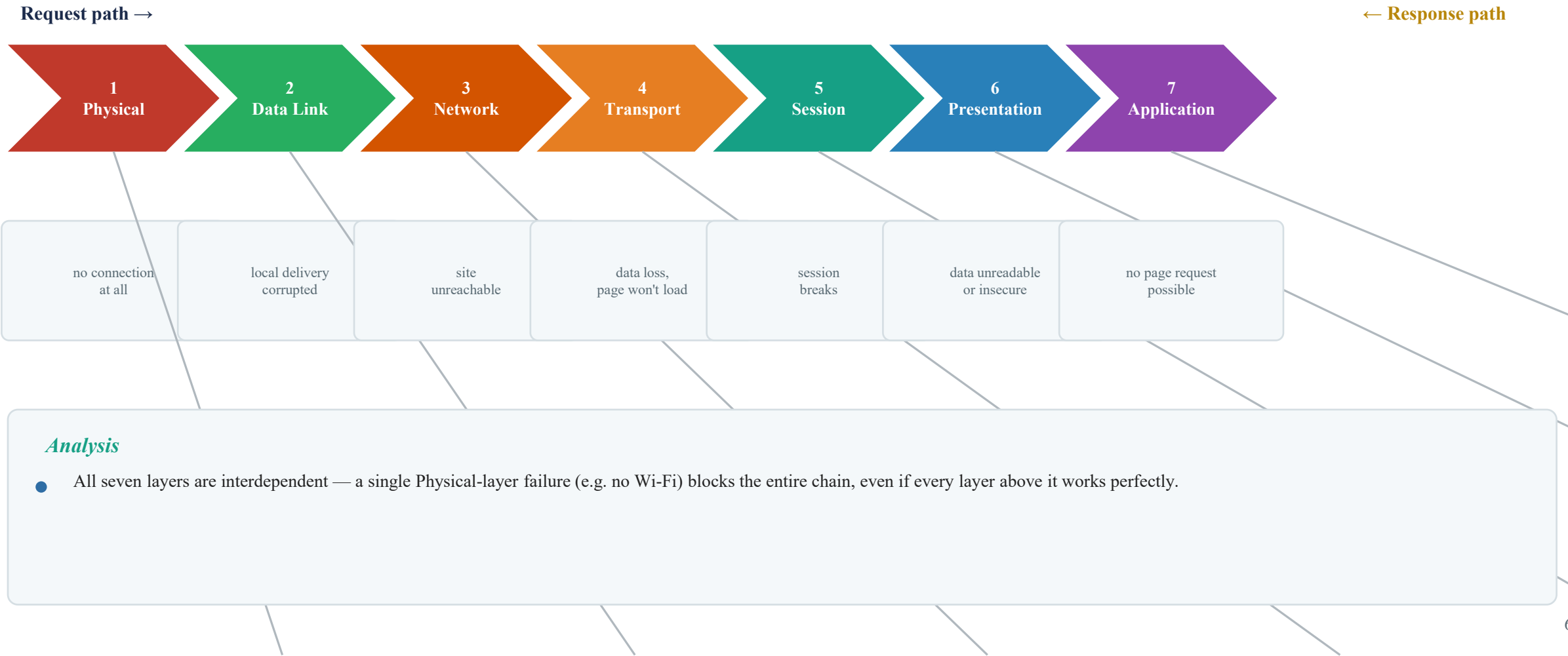
Analysis

- Data Link drops corrupted frames locally; Transport retransmits lost segments end-to-end — a two-tier safety net.

Question 5

Concept Map for Web Browsing Using OSI Layers

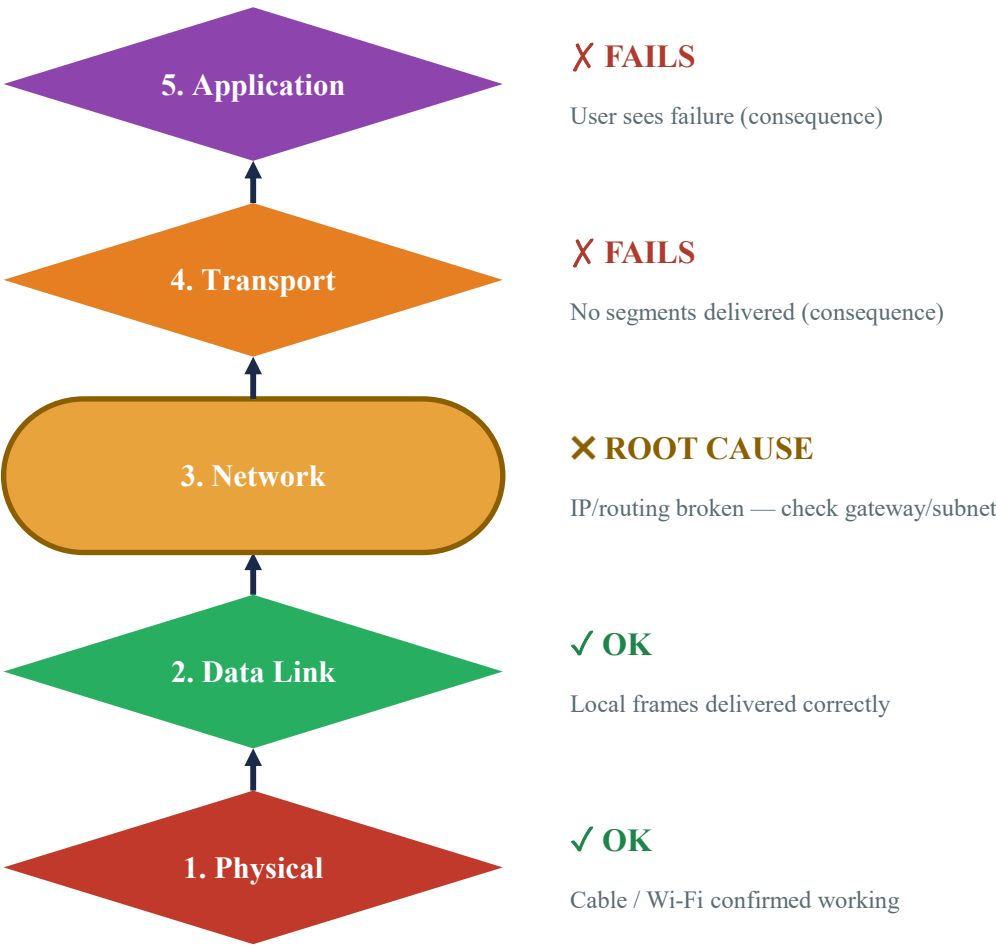
Request flows Layer 7→1 down, response flows 1→7 back up. Shape used: CHEVRON sequence



Question 6

Troubleshooting Using the Concept Map

Given: Physical ✓ → Data Link ✓ → Network X → Transport X → Application X | Shape used: DECISION DIAMOND + TERMINATOR



Root-cause logic

- Physical & Data Link confirmed OK
- Network is the first failing layer → root cause
- Transport & Application fail only as a consequence
- Bottom-up check narrows the fault fast